Please ensure this tracking sheet is completed as a group, as it will be used to monitor your individual participation within the team.

Team Members	Role/ Responsibilty	Email				
Igiraneza Nkuba Junior	Group Member	j.igiraneza1@alustudent.com				
Janviere Munezero	Group Member	j.munezero@alustudent.com				
Peggy Dusenge	Group Member	p.dusenge@alustudent.com				
Akinloye Emmanuel	Group Member	e.akinloye@alustudent.com				

Task Allocation and Tracker

	Task Allocated	Assigned Member	Deadline	Completion Status (Not Started/In Progress/Completed/Did not participate)	Reviewed by Team? (Yes/No)	Comments
	Part 3 Manual Iteration 1 + Part 4 (Gradient Descent code)	Akinloye Emmanuel	04/07/2026	Completed	Yes	Manual iteration 1 (output m=[-1.045,1.887], b=0.988); full Python implementation with SciPy derivative and both plots; MSE dropped 61→22
	Part 2 (Bayesian Probability) + Part 3 Iteration 2	Peggy Dusenge	04/07/2026	Completed	Yes	Bayes' theorem in pure Python on IMDb reviews, P(Positive|keyword) for 8 keywords; manual iteration 2 (output m=[-1.084,1.788], b=0.978)
	Part 3 Iteration 3 + PDF Compilation	Junior Nkuba	04/07/2026	Completed	Yes	Manual iteration 3 (output m=[-1.119,1.702], b=0.969); compiled handwritten calculations and contributions PDF for submission
	Part 1 (EM Algorithm) + Part 3 Iteration 4	Janviere Munezero	04/07/2026	Completed	Yes	EM Gaussian mixture from scratch with tracking table and live classification; manual iteration 4 (output m=[-1.149,1.627], b=0.962, MSE=22.09)

Meeting Attendance Log

Meeting Date	Agenda	Facilitator	Attendees	Key Discussion Points	Next Steps	

24-06-2026	Met to look through the assignment instructions	Emmanuel Akinloye	Emmanuel Akinloye, Peggy Dusunge, Junior Nkuba, Janviere Munezero	Reviewed the assignment instructions together and broke it into four parts (EM algorithm, Bayesian probability, gradient descent, and PDF compilation). Agreed on shared conventions for the gradient descent: learning rate $\alpha=0.001$, a single scalar bias b, and MSE as the cost function.	Each member to begin their assigned part and bring initial progress to the next meeting.	
26-06-2026	Shared task and gave deadlines for tasks	Peggy Dusunge	Emmanuel Akinloye, Peggy Dusunge, Junior Nkuba, Janviere Munezero	Allocated specific tasks to each member and set individual deadlines. Confirmed the manual gradient descent iteration chain (iteration 1 → 2 → 3 → 4) and who owns each iteration so the handoff numbers line up.	Members to complete first drafts of their parts; gradient descent iterations to be done in sequence.	
01-07-2026	Follow up on deliverables from team	Junior Nkuba	Emmanuel Akinloye, Peggy Dusunge, Junior Nkuba, Janviere Munezero	Followed up on each member's deliverables. Cross-checked the manual gradient descent results against the Python code output and confirmed they match (MSE falling 61→47→36→28→22). Reviewed the EM tracking table and the Bayesian keyword results.	Fix any remaining bugs, align handwritten iterations with the code, and begin assembling the Jupyter notebook and PDFs.	
03-07-2026	Final review and submission	Janviere Munezero	Emmanuel Akinloye, Peggy Dusunge, Junior Nkuba, Janviere Munezero	Final review of all four parts as a group. Confirmed the notebook runs end to end, the manual PDF is compiled, and the README and contributions documents are complete. Verified all deliverables are consistent across the sheet, code, and handwritten work.	Compile final submission package; each member to prepare talking points for their section of the presentation.	

05-07-2026	Preparation for presentation	Akinloye Emmanuel	Emmanuel Akinloye, Peggy Dusunge, Junior Nkuba, Janviere Munezero	Rehearsed the 15-minute presentation and confirmed each member speaks on every part. Practiced likely Q&A (why not split at the mean in EM, why SciPy's gradient matches the manual derivation, keyword-choice rationale).	Submit the assignment and present in the booked slot.	